LLM used: GitHub Copilot (VS Code)

Prompt 1:

User:
Execute the main.py file and tell me what bugs you observe.

Copilot:
I ran the game entry point with Python, and the program does not crash, but it does exhibit a serious gameplay bug:

- Command run: python .\main.py
- Observed output:
    - Game over! Final score: 0
- Exit status: 0 (clean exit, but the game already ended)

What I observed:

- The game reaches a loss state almost immediately, before the player has a realistic chance to react. That is a logic bug in the game loop / collision flow, not a Python exception.
- In game/game_engine.py, the engine:
    - increases self.speed every frame with no ceiling,
    - spawns obstacles over time,
    - then checks collisions after moving the obstacle, which can allow hits to be missed or trigger bad state transitions unexpectedly.
- The file even contains comments warning about the collision bug and the speed runaway behavior.

So the main issue is: the game is effectively “ending on startup” / very early, which means the core gameplay logic is broken even though the script itself runs successfully.

Prompt 2: Fix bugs and implement all tasks

User:
Fix the bugs, and also add the following features:

Task 1: Refine Collision Detection
The game speeds up forever with no limit, and once it's fast enough obstacles can zip past the player without the hit ever registering. Investigate and enhance collision accuracy (and/or the speed ramp) so it stays fair no matter how long a run lasts.

Task 2: Implement Game Over Condition
Add a screen that displays the final score once the player collides with an obstacle, then gracefully waits for input instead of just printing to the console.

Task 3: Add Replay Option
After Game Over, allow the user to play again by choosing a difficulty (Easy, Medium, or Hard starting speed/spawn rate), or exit.

Task 4: Add Sound Feedback
Add basic sound effects for jumping, passing an obstacle (scoring), and the game-over moment.

Copilot:
I'll trace the current game state flow, input handling, and available assets/tests first, then implement the game-over/replay flow and verify the collision and sound behavior.

I found the main gaps: collision checks only sample the post-move frame, the speed is uncapped, and game-over only prints to stdout. I'm replacing those with swept collision checks plus a speed ceiling, a keyboard-driven replay screen, and generated sound effects so no external audio assets are required.

The core changes are in place. The targeted checks passed for fast-crossing collision detection, capped speed, replay selection/reset, and clean startup/exit; I'm doing a final pass for the rendered game-over screen and all three sound-trigger paths.

The sound trigger and rendering checks also pass. I'm making the replay choices show their actual starting speed and spawn interval on-screen, so the difficulty selection is clear rather than just labeled.

Implemented the game fixes and requested features.

- game/game_engine.py: Added swept collision detection to catch obstacles that cross the player between frames, capped speed at 12, and added difficulty-specific starting speeds and spawn intervals.
- main.py: Added handling for replay difficulty selection and exiting.
- game/sound_effects.py: Added synthesized jump, score, and game-over sounds.
- README.md: Documented controls and difficulty settings.

After a collision, the game now displays the final score and offers **1 — Easy**, **2 — Medium**, **3 — Hard**, or **Esc — Exit**. Audio-device failure won't prevent the game from running.

Verified: headless engine checks covered fast-crossing collisions, the speed cap, sound triggers, game-over rendering, replay/reset, and exit handling. The main.py startup/exit smoke test passed, as did git diff --check.

Prompt 3: Execute and verify

User:
Execute the files and verify that everything works as intended.

Copilot:
All execution checks passed. I ran main.py with Pygame's dummy video/audio drivers; it initialized and exited cleanly. I also executed the game modules and checked:

- Fast-crossing collisions are detected and speed stays capped.
- Difficulty selection, replay reset, and exit handling work.
- The game-over screen renders with the final score.
- Jump, score, and game-over sound effects can be created and played.
- git diff --check reported no whitespace errors.